Get started with the Tripo API in 15 minutes and generate your first 3D model.

1 Get an API Key

Go to the **API Keys** page, create a new key, and copy it. Store it securely — it is shown only once.

> **Security tip:**
>
> Store your API key in an environment variable (export TRIPO_API_KEY="sk-..."). Never hard-code it in source code or commit it to repositories.

2 Send Your First Request

Generate a 3D model from a text description. All generation APIs are asynchronous — the response returns a task_id immediately.

cURL Python JavaScript Copy

```
-cmd">curl -X POST https://openapi.tripo3d.ai/v3/generation/text class
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer " \
  -d '{
    "prompt": "a cute cat",
    "model": "v3.1-20260211"}'
```

200 Response — 200 OK

```
{
  "code": 0,
  "data": {
    "task_id": "task_abc123"}
}
```

3 Query the Task Result

Use task_id to poll until status: "success". Poll every 2 seconds. Typical generation takes 10–120 seconds.

cURL Python JavaScript Copy

```
-cmd">curl -X GET https://openapi.tripo3d.ai/v3/tasks/{task_id} \
  -H "Authorization: Bearer "
```

200 Response — Task Complete

```
{
  "code": 0,
  "data": {
    "task_id": "task_abc123",
    "type": "text_to_model",
    "status": "success",
    "progress": 100,
    "output": {
      "model_url": "https://...",
      "rendered_image_url": "https://..."}
  }
}
```

4 Download the Model

Download the GLB 3D model file from output.model_url in the response. You can use it directly in:

Blender | Unity | Unreal Engine | Three.js | Babylon.js | model-viewer

Note:

Model URLs expire after 5 minutes. Download immediately after the task succeeds. Use output.model_url for the 3D model file.

Advanced Workflows

Explore complete API pipelines for specific use cases and features.

By Scenario | By Feature

Game-Ready Character

Generate a low-poly character with skeleton and animations, ready for Unity/Unreal.

High-Fidelity Asset

Generate the highest quality PBR model for product visualization or cinematic rendering.

3D Printing

Generate a pure geometry mesh without textures, export as STL for slicing and printing.

Animated Scene

Create characters with different animations for game levels or short films.

E-commerce Product

Turn a product photo into an interactive 3D model for web AR try-on or product pages.

Multiview Pipeline

Use multiple reference views for the most accurate 3D reconstruction.

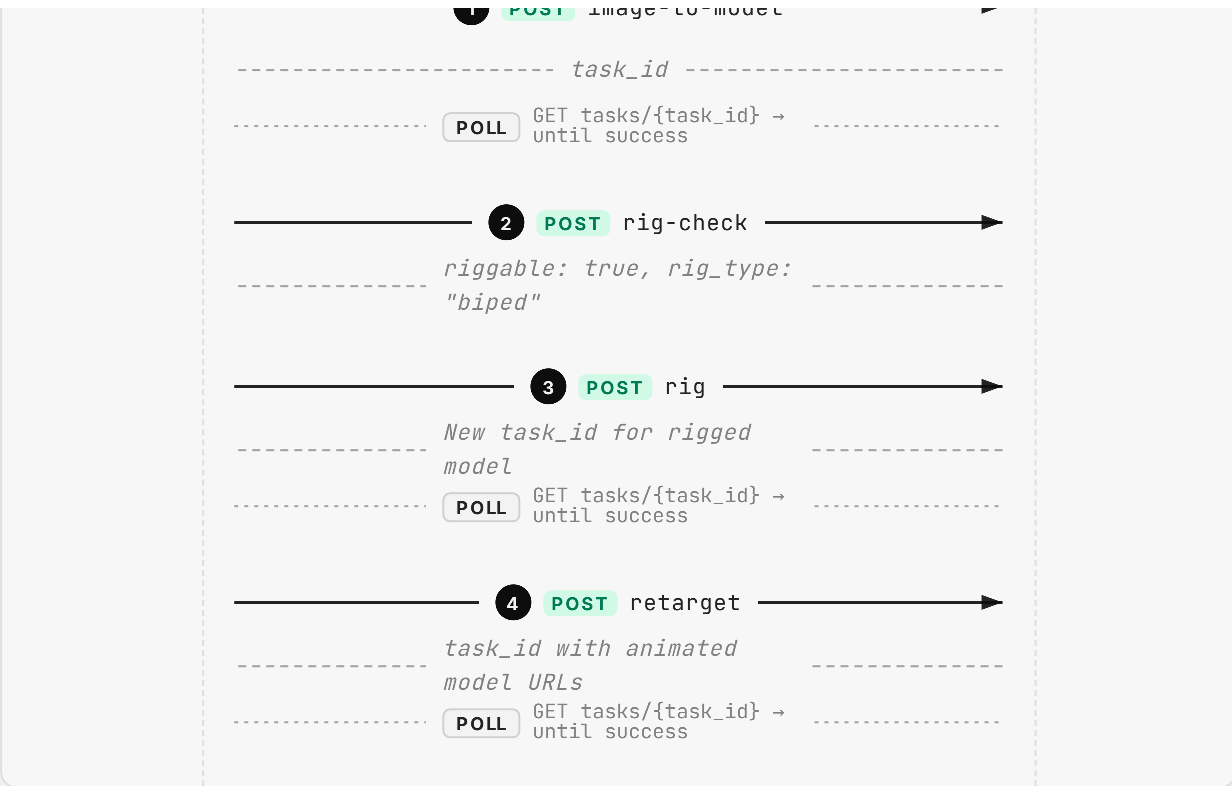
Game-Ready Character

Generate a low-poly character with skeleton and animations, ready for Unity/Unreal.

Call Sequence

Your App | Tripo API

1 POST image-to-model



API Data Flow

POST `image-to-model` → POST `rig-check` → POST `rig` → POST `retarget`

1 POST /v3/generation/image-to-model

INPUT

Image URL or file_token

OUTPUT

task_id

→ Poll GET /v3/tasks/{task_id} until status=success

2 POST /v3/animations/rig-check

INPUT

```
{ input: "task_abc123" }
```

OUTPUT

```
riggable: true, rig_type: "biped"
```

→ If riggable=true, pass task_id + rig_type to rig

3 `POST /v3/animations/rig`

INPUT

```
{ input: "task_abc123", rig_type: "biped" }
```

OUTPUT

```
New task_id for rigged model
```

→ Poll until success, pass rigged task_id to retarget

4 `POST /v3/animations/retarget`

INPUT

```
{ input: "task_rig456", animations: [...] }
```

OUTPUT

```
task_id with animated model URLs
```

→ Download output.model_urls

Key Parameters

PARAMETER	VALUE	WHY
`model`	`P1-20260311`	Best low-poly topology for game assets

PARAMETER	VALUE	WHY
`face_limit`	`5000`	Mobile/game-friendly polygon count
`texture`	`true`	Visual fidelity for characters
`rig_type`	`biped`	Humanoid skeleton for character
`spec`	`mixamo`	Compatible with Unity/Unreal import
`animations`	`preset:walk` `preset:idle` `preset:run`	Essential locomotion set

ESTIMATED TIME

~105s

COST (CREDITS)

~85

cURL Python JavaScript Copy

```python
import requests, time

HEADERS = {"Authorization": "Bearer ", "Content-Type": "application/json"}

def poll(task_id):
    while True:
        r = requests.get(f"https://openapi.tripo3d.ai/v3/tasks/{task_id}",
        if r["data"]["status"] == "success": return r["data"]
        if r["data"]["status"] in ("failed","cancelled"): raise Exception(r
        time.sleep(2)

# Step 1 — Generate model
task_id = requests.post("https://openapi.tripo3d.ai/v3/generation/image-to-
    headers=HEADERS, json={"file_token":"","model":"P1-20260311","face_limi
).json()["data"]["task_id"]
poll(task_id)
```

Developer Tips

- Always call rig-check before rig — it returns the recommended rig_type and prevents failures
- Poll tasks every 2 seconds; do not exceed 1 request/second to avoid rate limits
- Use spec: "mixamo" for Unity/Unreal; use spec: "tripo" for custom pipelines

Common Pitfalls

- If rig-check returns riggable=false, try regenerating with a cleaner prompt (avoid complex poses)
- Retarget only accepts rigged model task_ids — passing a raw generation task_id will fail